Lab - Imaging a Directory Using FTK Imager

Overview

Digital forensic analysis reviews and investigates data collected through digital communications and computer networks. The objective is to recover, analyze, and present digital or computer-based data so they can be used as evidence in court.

Lab Requirements.

- One Windows 10/11 installation (virtual or otherwise).
- One installation of FTK Imager.
- One directory with digital evidence to collect.

Begin the lab!

Task 1: FTK Imager – Imaging a Directory

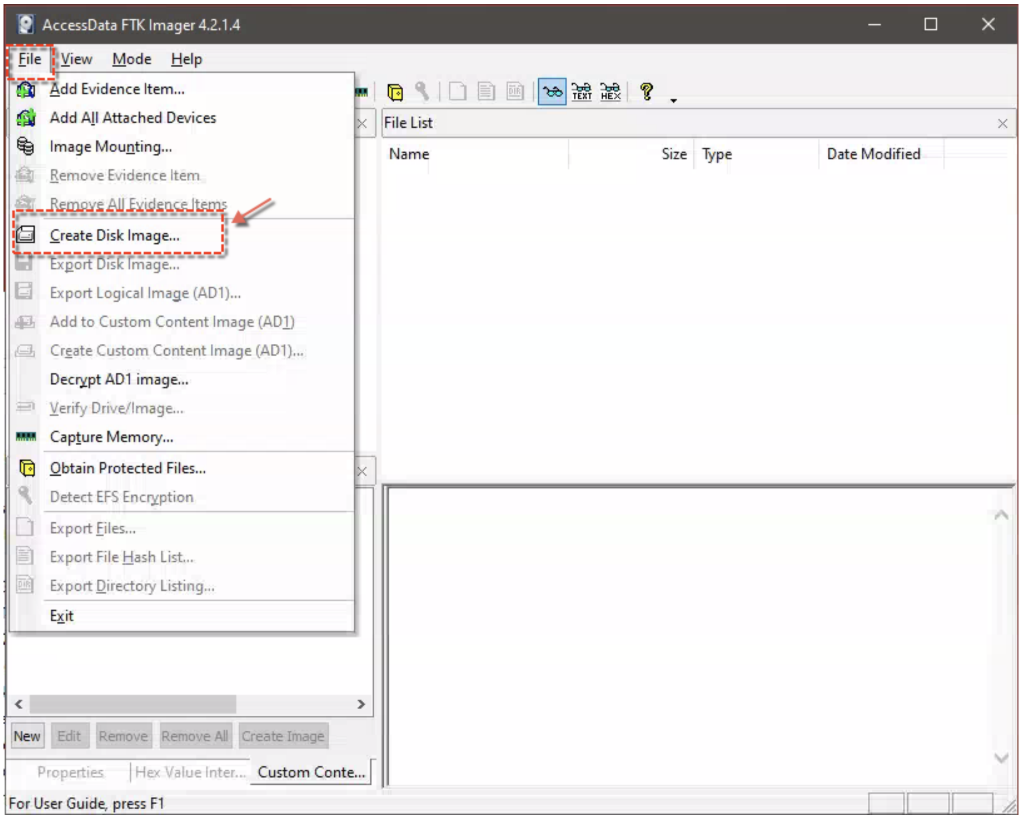
In this lab, you will use FTK Imager to image a directory folder. Usually, investigators will image (i.e., make a bit-by-bit copy) a whole drive to make a duplicate. In this exercise, you will only image a single directory and its contents.

In the following example of experimenting with FTK Imager, keep in mind that you will only select to image a disk directory, choose the directory in the tool's interface options, and then select the output of that image. You could then mount the resulting image and explore its contents to verify that it was, in fact, a captured image of the directory you had specified.

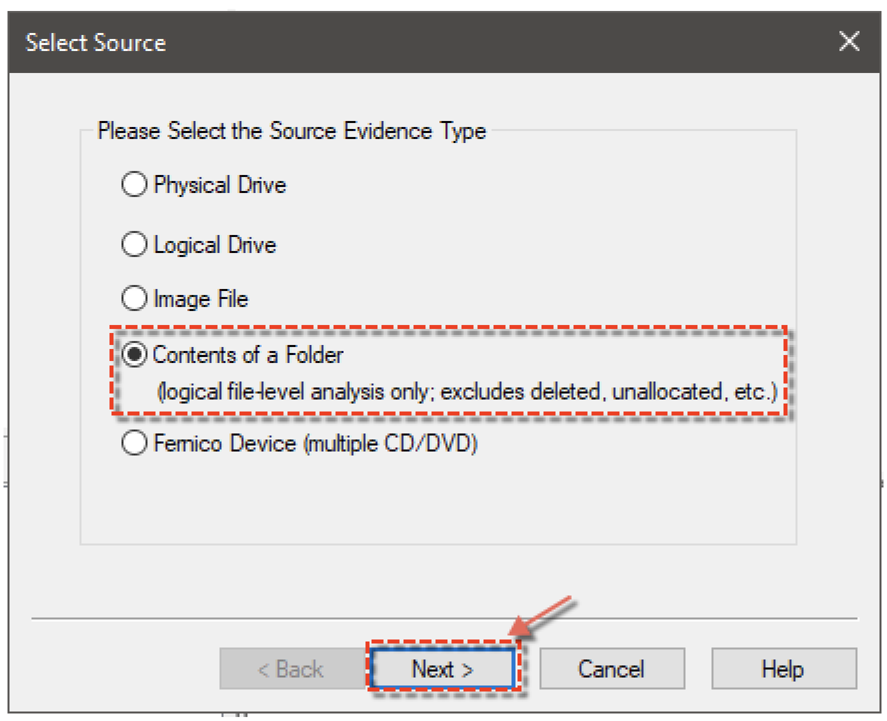
Image Creation Process

In this task, you are required to select an image from a disk directory, select the output of that image, create a hash of the image, review the summary, and get it ready for analysis.

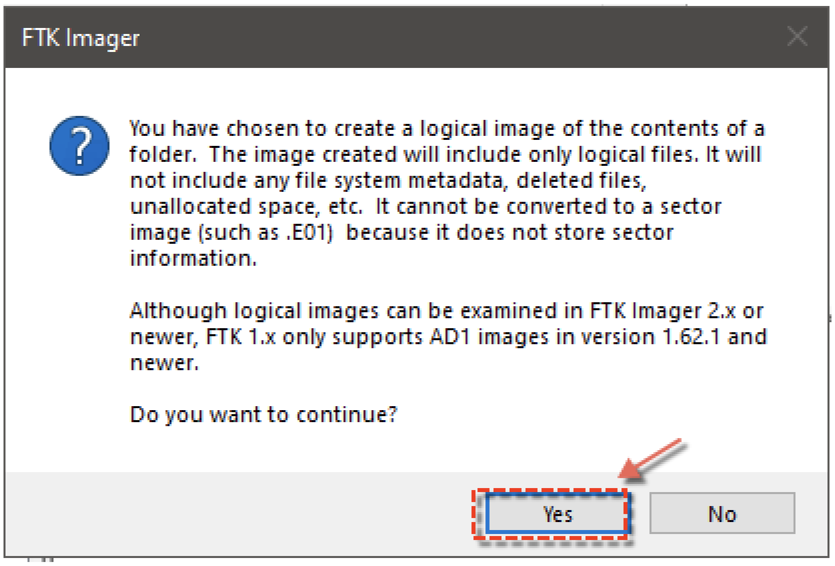
1. Launch FTK Imager
2. Once the program starts, select **File,** and click Create Disk Image from the drop-down menu.



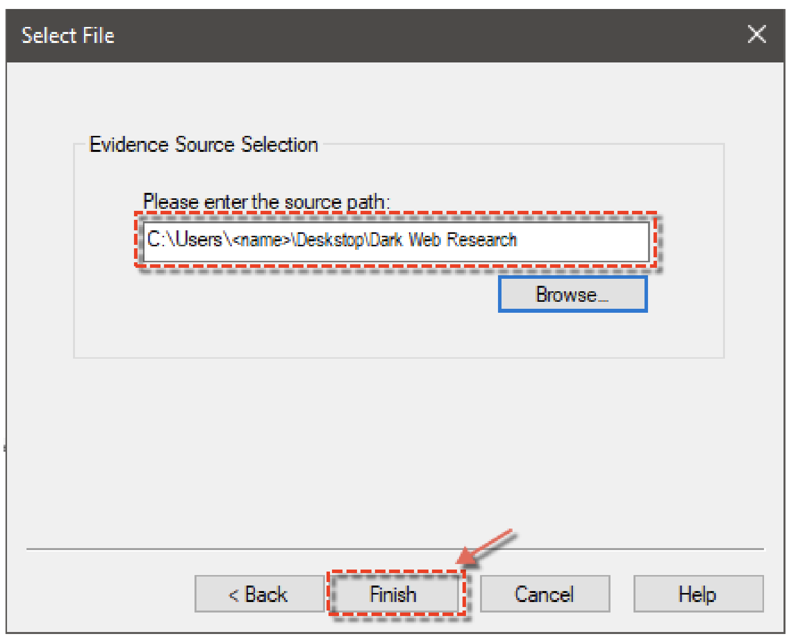
3. Select the Contents of a Folder option in the Select Source dialog box. In this exercise, you are going to create a bit-by-bit image of a folder due to size and time constraints. Click **Next**.



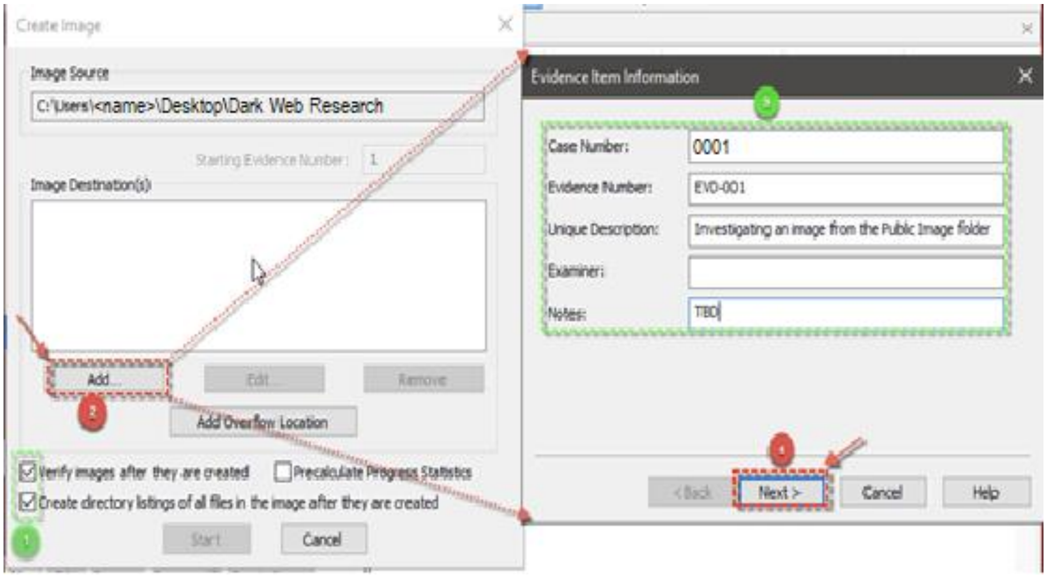
4. Notice the information in the pop-up message. Review it and click **Yes** to continue.



5. You will create an image of the **Dark Web Research** folder located on your desktop, as shown below. Browse to select the folder and click **Finish**.

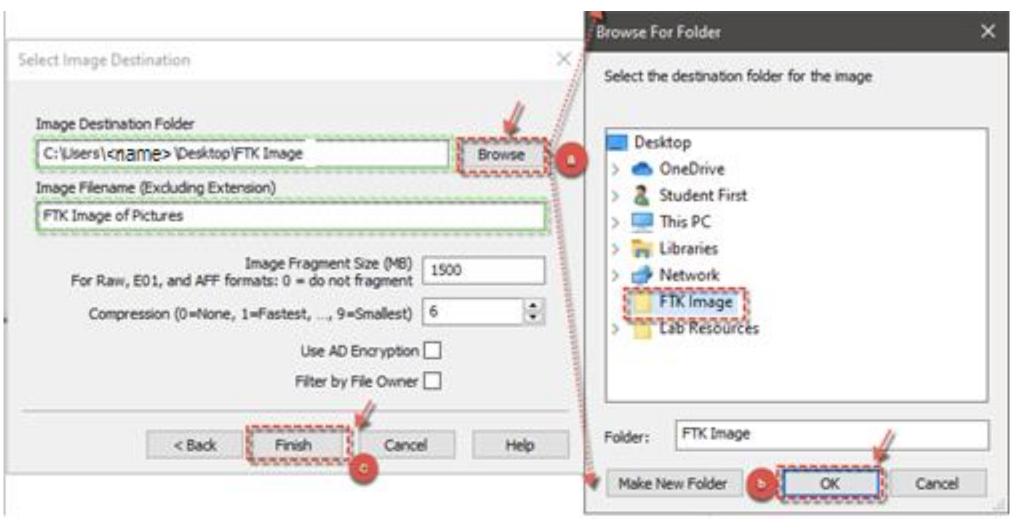


6. In the **Image Source** dialog box, make sure that you select boxes for the **Verify images after they are created** and **Create directory listings of all files in the image after they are created**. Then click **Add** to fill out the **Evidence Item Information** pop-up window as you wish. Finally, click **Next**.

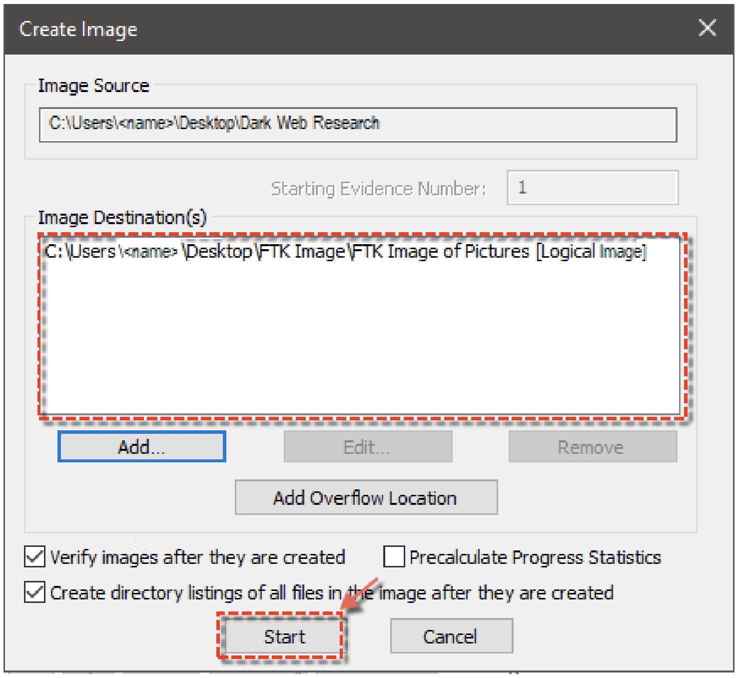


Note: Before the next step, you must create an **FTK Image** folder on your desktop. Follow the steps below to select the image for your analysis.

7. This step requires you to select a destination path in the **Select Image Destination** dialog box to create hashes of the files. Follow these steps:
    a. First, make sure you provide the image name in the **Image Filename** field. Then, click the **Browse** button.
    b. Next, select the **FTK Image** folder from your desktop and click **OK** to select the folder.
    c. After selecting the image, click **Finish**.

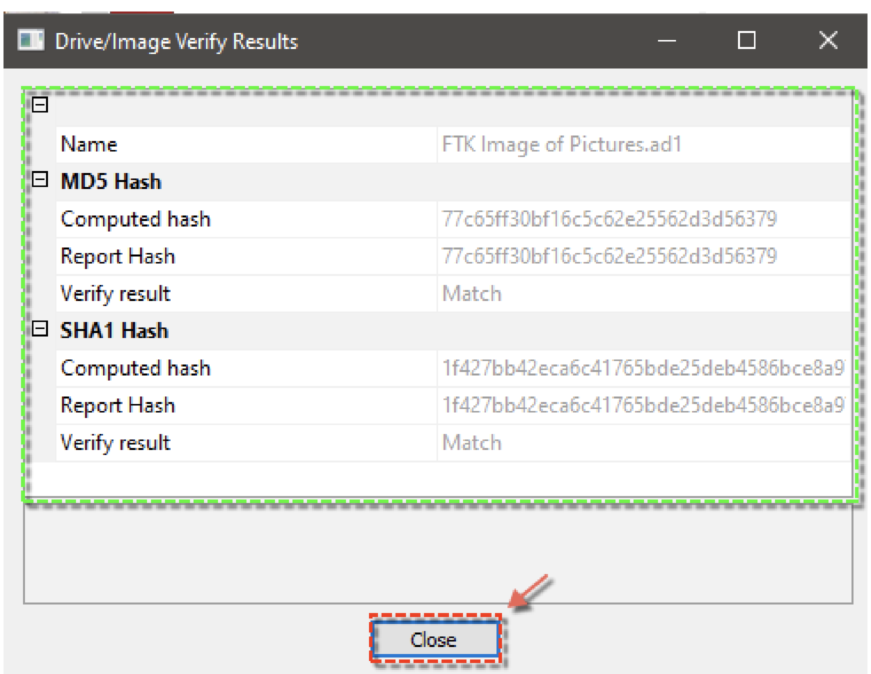


8. This brings you back to the **Create Image** window with the **Image Destination(s)** box populated. Click **Start** to start the image creation process.

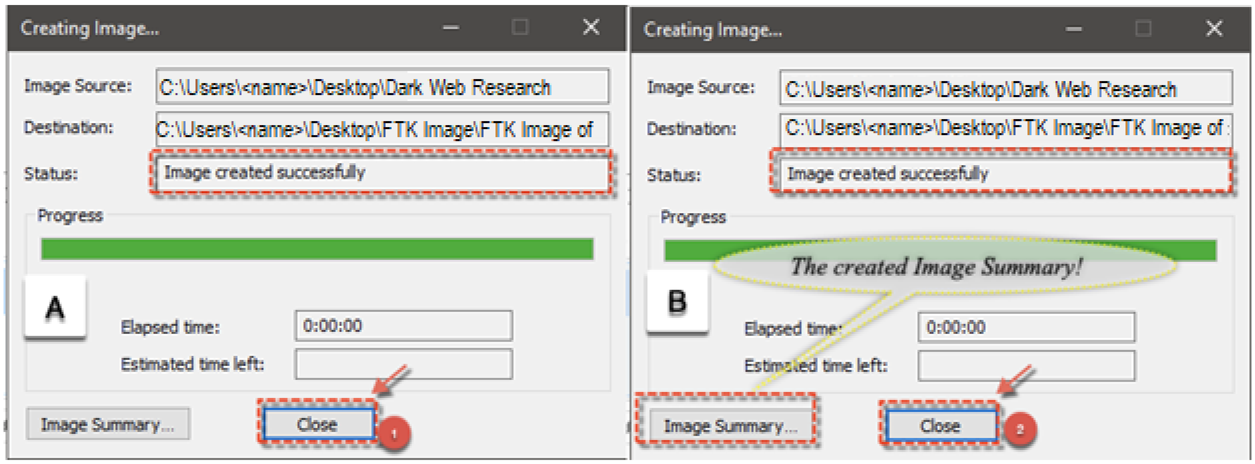


Note: The image creation starts, and a dialog box (**Drive/Image Verify Results**) will pop up, indicating the hashes that have been created. The hashes are the values that ensure the integrity of the image. Later, possibly during litigation, these hash values could be used to prove that the image was not altered after the time of initial capture.

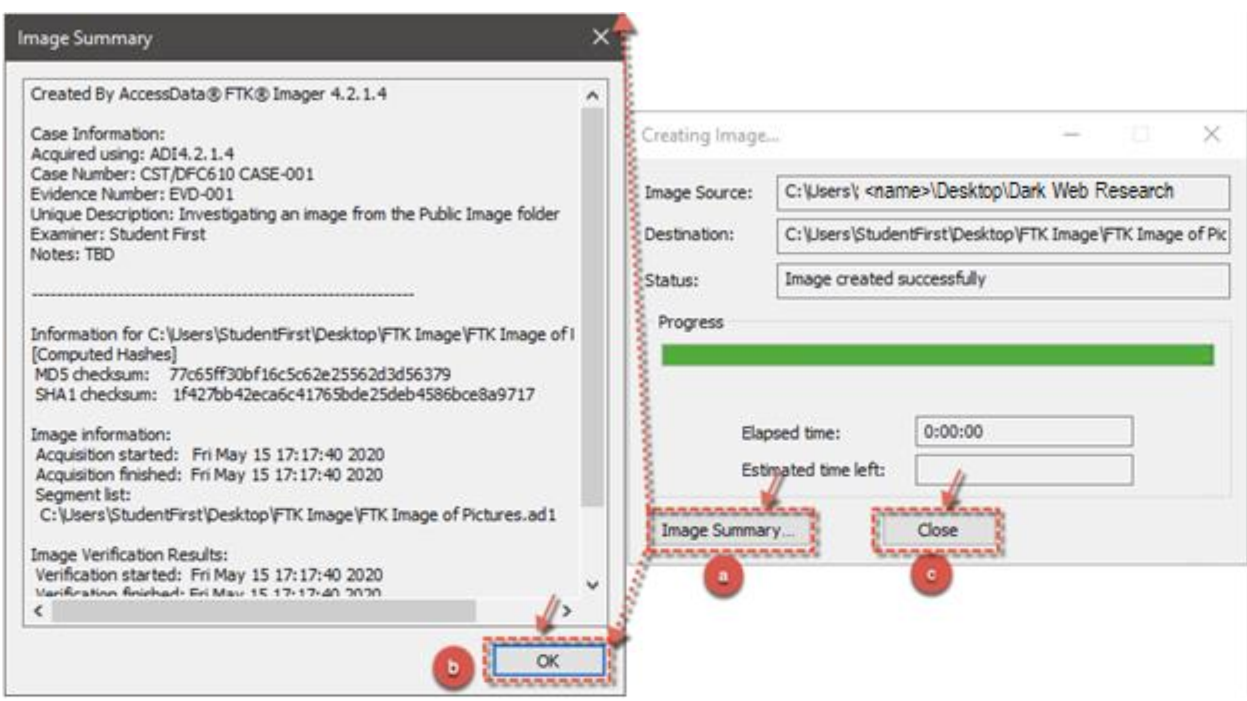
9. Click **Close**.



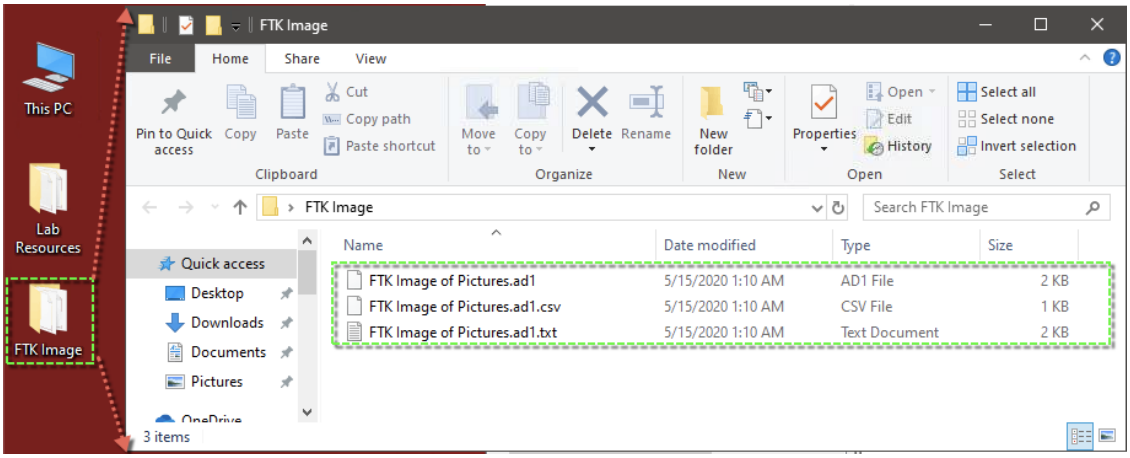
10. Clicking the **Close** button from above creates the **Creating Image**... dialog box below (**Box A** on the left). Notice that the **Status** field shows that the directory listing has been created successfully. Clicking **Close** again creates the **Image Summary** button (**Box B** on the right).



11. Next, complete the following steps.
    a. First, click the **Image Summary** button.
    b. Then, review the details in the **Image Summary** pop-up window and click **OK** when done. What can you determine from the summary?
    c. Finally, click **Close**.



12. This ends the image-creating process. The images can be found in the **FTK Image** folder on your desktop. Double-click the folder to review your image.



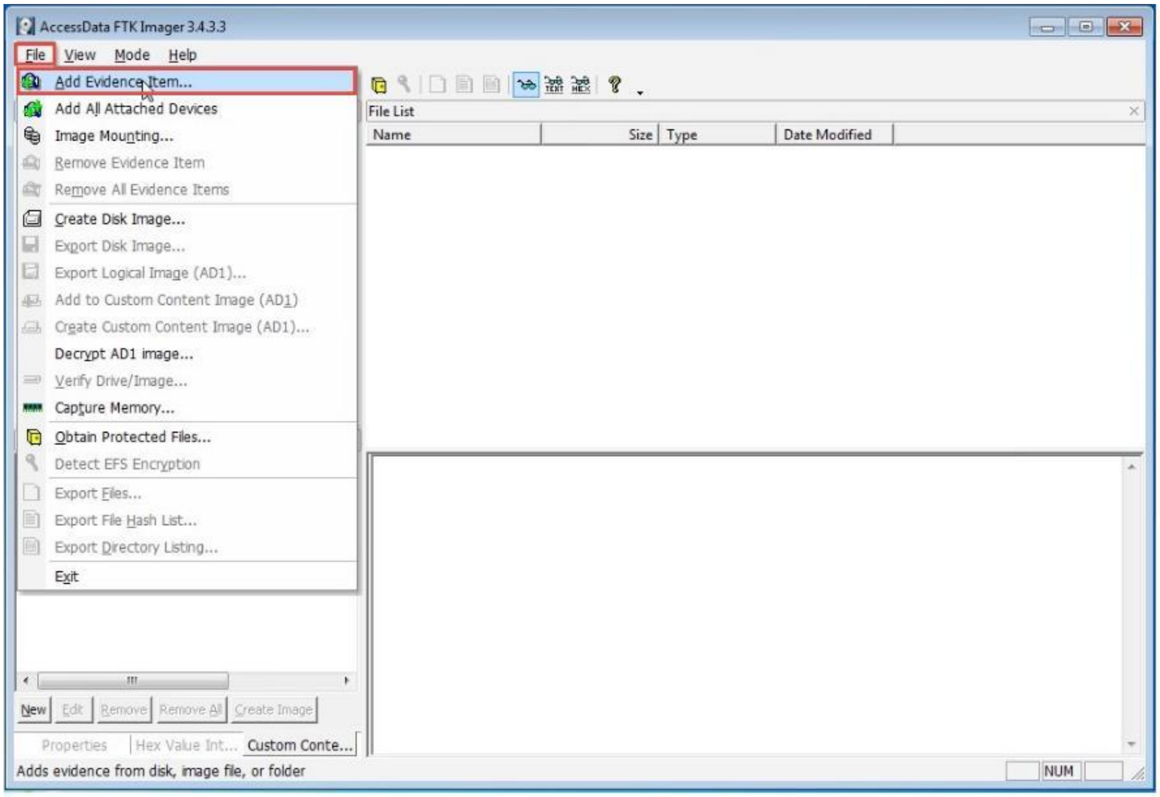
Task 2: Image Verification and Forensic Analysis

You will now conduct verification and forensic analysis of your created image. You will mount the resulting image and explore its contents to verify that it was, in fact, a captured image of the directory you had specified.

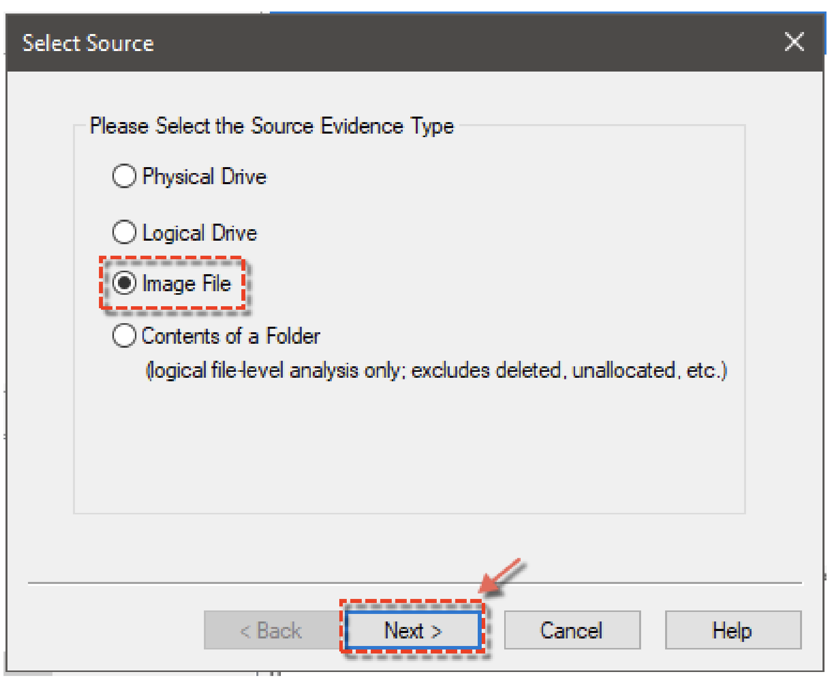
Image Verification

You will read the image back into FTK Imager to verify the image is indeed stored. In a real investigation, the above steps are performed to capture the information of a system that is being investigated. The image is usually stored in a separate storage device such as a thumb drive or an external hard drive. A forensic investigator then loads the evidence into a separate system for analysis.

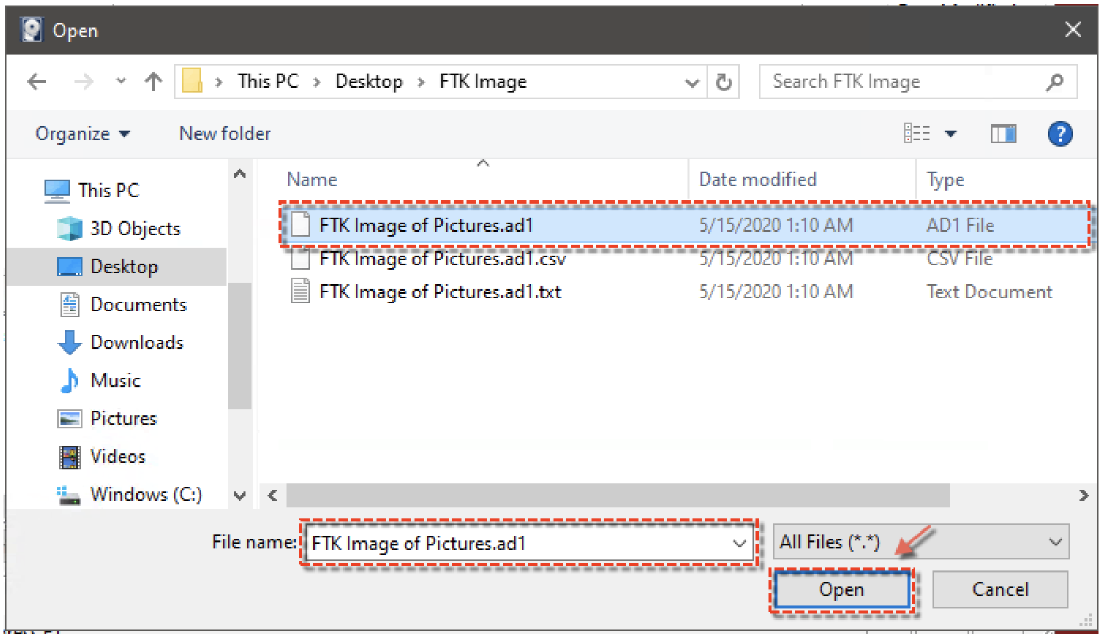
1. To load your copy using FTK Imager, click **File** > **Add Evidence Item...** in **FTK Imager**.



2. In the **Select Source** window, choose **Image File** (i.e., where the FTK image was stored) and click **Next**.

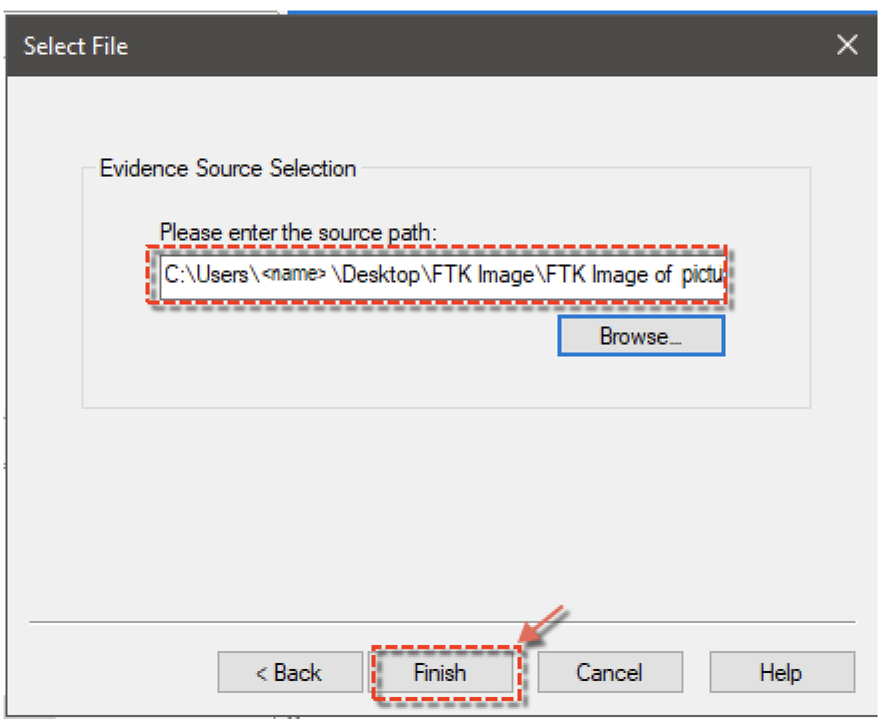


3. Select the file location where the FTK image was stored and click **Open**.

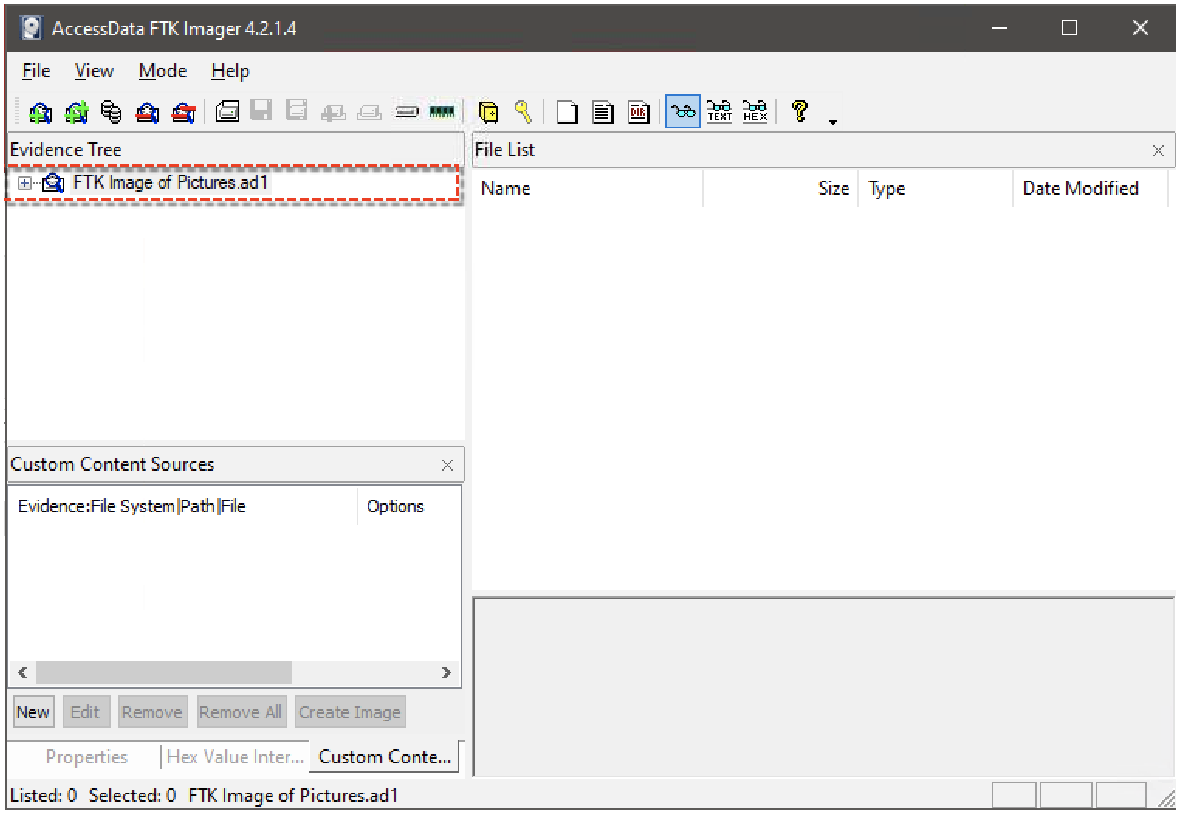


Selecting an Image File

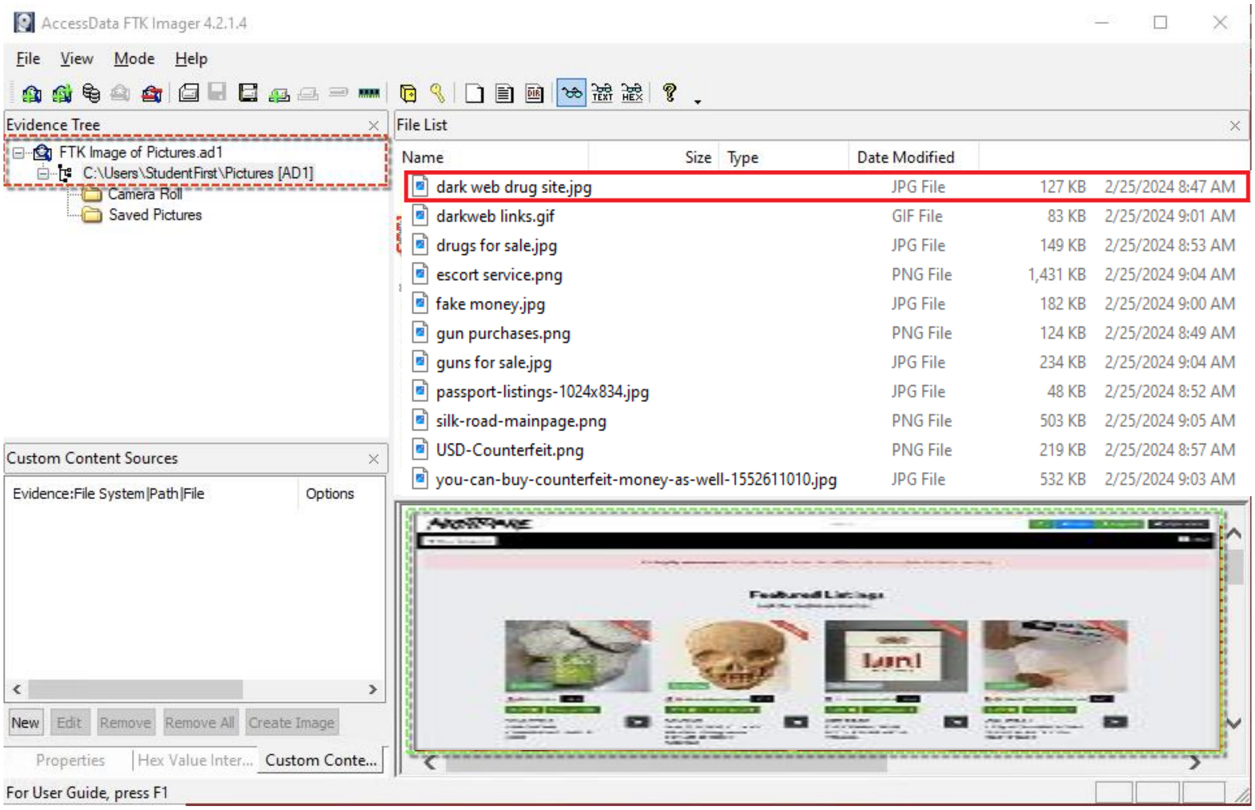
4. After selecting the **FTK Image of Pictures.ad1** file, click **Finish** to have it added to **FTK Imager** as an evidence item.



5. Next, you should see the **FTK Image of Pictures.ad1** file in the **Evidence Tree** window with the plus (+) sign.



6. Expand the **FTK Image of Pictures.ad1** by clicking the plus (+) sign to see the files contained in the **Pictures** folder. Click the first image file and any other image in the **File List** pane to see the images in the preview window. Notice that these image files should be the exact copies of the original files before the image was created.



Summary

In this lab, you learned some of the concepts associated with forensic imaging and analysis.

For a digital forensic investigator, the next step after obtaining a forensic image of the evidence would be to analyze the evidence using FTK, EnCase, or Autopsy.

End of the lab!